

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

ME3100-1 – DESIGN OF MACHINE ELEMENTS

(For ME) *

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. The ratio of the ultimate stress to the design stress is known as _____.
A) elastic limit
B) strain
C) factor of safety
D) bulk modulus
2. Principal plane is a plane on which the shear stress is _____.
A) zero
B) varying
C) maximum
D) constant
3. Maximum principal stress theory is applicable for _____ materials.
A) ductile
B) brittle
C) elastic
D) viscous
4. Maximum shear stress theory is applicable for _____ materials.
A) ductile
B) brittle
C) elastic
D) viscous
5. One of the assumptions made in deriving the stress equation in curved beams is that, the _____ is the same in tension as well as in compression.
A) modulus of plasticity
B) modulus of rigidity
C) modulus of elasticity
D) Poissons' ratio
6. Stress distribution in curved beams is _____ in nature.
A) parabolic
B) hyperbolic
C) linear
D) cubic
7. In the case of curved beams, the _____ of the section is shifted towards the centre of curvature of the beam.
A) centroidal axis
B) neutral axis
C) line of action
D) Outer radius
8. The maximum bending stress in a curved beam having symmetrical section always occurs at the _____.
A) centroidal axis
B) neutral axis
C) inside fibre
D) outside fibre
9. Which of the following types is not fillet weld?
A) butt joint
B) lap joint
C) T-joint
D) corner joint
10. The length of each of the two equal sides of a parallel fillet weld is called _____.
A) leg
B) throat
C) arm
D) web
11. Strength of parallel fillet weld is _____ than strength of transverse fillet weld.
A) lesser
B) greater
C) same as that of
D) zero

12. What kind of stresses does a welded joint undergo?
 A) tensile stress B) compressive stress
 C) Direct shear and torsional shear stress D) bearing stress
13. The friction material of the clutch should have _____
 A) high coefficient of friction B) low coefficient of friction
 C) high surface hardness D) high endurance limit strength
14. In case of multi-plate clutch if n_1 is number of disks on driving shaft and n_2 is the number of disks on driven shaft, then the number of pairs of contacting surfaces is given by _____
 A) $n_1 + n_2$ B) $n_1 + n_2 + 1$
 C) $n_1 + n_2 - 1$ D) $n_1 + n_2 - 2$
15. The friction material of the brake should have _____
 A) high coefficient of friction B) low coefficient of friction
 C) high surface hardness D) high endurance limit strength
16. In case of a new clutch, the uniform _____ theory is applicable.
 A) pressure B) wear
 C) normal force D) axial force
17. The size of a gear is usually specified by _____.
 A) Pressure angle B) Pitch circle diameter
 C) circular pitch D) diametral pitch
18. Which of the following statement is incorrect for gears?
 A) The pitch circle diameter is equal to the sum of module and number of teeth B) The addendum is less than the dedendum
 C) The pitch circle is always greater than the base circle D) Circular pitch is expressed as πm , where m is the module in mm.
19. Lewis equation is applied _____.
 A) only to the pinion B) only to the gear
 C) to stronger of the pinion or gear D) to weaker of the pinion or gear
20. If T is the actual number of teeth on a helical gear and β is the helix angle for the teeth, then $T \sec^3 \beta$ gives _____.
 A) formative number of teeth B) normal module
 C) dynamic strength D) wear strength

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I.

Marks	BT*	CO*	PO*
8	L*1	1	1

1. a) Explain the various types of designs.
 b) A crane hook has a trapezoidal cross section with the following data: Inside width 87.5 mm, Outside width 25 mm, Depth 112.5 mm. The line of action of the load passes through the center of curvature. The radius of curvature of inner side is 62.5 mm. Evaluate the maximum stresses developed under a load of 90 kN.
2. a) A steel member is loaded as shown in Figure. Q2(a). Determine the magnitude of (i) Maximum normal stress (ii) Minimum normal stress (iii) Maximum shear stress. Interpret the results obtained.

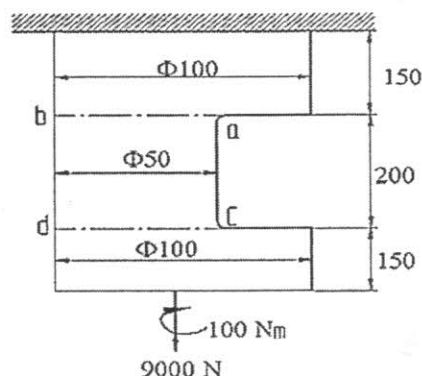


Figure. Q2(a)

8	L3	1	2
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- b) Arrive at the Soderberg's equation when a machine member is subjected to fatigue loading. Also, write the modified Soderberg's equation considering the modifying factors for endurance limit.

8 L1 2 1

3. a) Determine the diameter of the hole in Figure. Q3(a), if the stress concentration factor at the hole is to be the same as at the fillet.

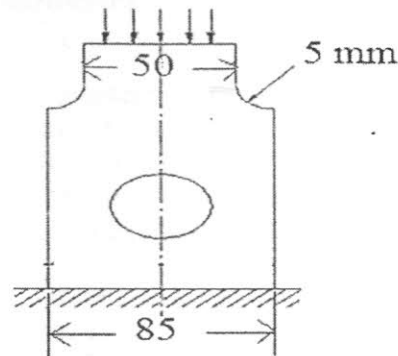


Figure. Q3(a)

8 L2 1 2

- b) A steel shaft is subjected to a bending moment which varies from 100Nm to 200Nm and transmits 10kW at 150 rpm. The torque varies over a range of $\pm 40\%$. The shaft is made of steel whose yield stress = 400 N/mm² and endurance stress = 300 N/mm². Surface coefficient factor=0.9, size factor=0.85, factor of safety=2.5, stress concentration factor = 1.94. Determine the diameter of the shaft for infinite life.

8 L3 2 2

Unit – II

4. a) Determine the size of weld required for the joint shown in Figure. Q4(a), if the allowable shear stress in the weld is limited to 80 MPa.

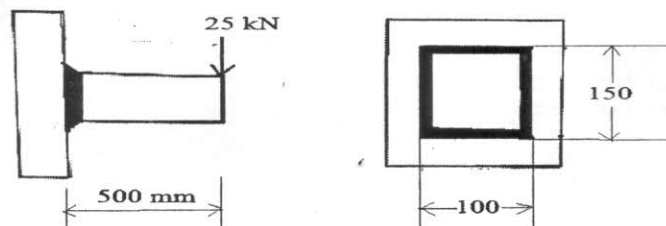


Figure. Q4(a)

10 L3 3 2

- b) A multiple plate clutch has steel on bronze is to transmit 8 kW at 1440 rpm. The inner diameter of contact is 80 mm and outer diameter of contact is 140 mm. The clutch operates in oil with expected coefficient of friction of 0.1, the maximum allowable pressure is 0.35 MPa. Assume uniform wear theory and determine the following (i) Number of steel and bronze plates (ii) Axial force required (iii) Actual maximum pressure.

6 L2 4 2

5. a) Design a valve spring for an automobile engine. When the valve is closed, the spring produces a force of 45N and when it opens, produces a force of 55N. The spring must fit over the valve bush which has an outside diameter of 20 mm and must go inside a space of 35 mm. The lift of the valve is 6 mm. The spring index is 12. The allowable stress may be taken as 330 MPa. Modulus of rigidity is 80GPa.

10 L3 3 2

- b) A cone clutch has a semi cone angle of 12° to transmit 10 kW at 750 rpm. Width of the face is 25 % of the mean diameter of friction lining. If the normal intensity of pressure between contacting surface is not to exceed 0.85 bar, assuming uniform wear, coefficient of friction = 0.2, calculate the clutch dimensions. Also find the axial force while running/at the beginning of engagement.

6 L2 4 2

6. a) A Belleville spring is made of 3 mm thick sheet steel with an outside diameter of 125mm and an inside diameter of 50mm. The spring is dished 5mm. The maximum stress is to be 500 MPa. Determine
 (i) Safe load carried by the spring
 (ii) Deflection at this load
 (iii) Stress produced at the outer edge
 (iv) Load for flattening the spring
 Assume poisson's ratio = 0.3 and $E = 200 \text{ GPa}$
- b) A simple band brake of drum diameter 600 mm has a band passing over it with an angle of contact of 225° , while one end is connected to the fulcrum, the other end is connected to the brake lever at a distance of 400 mm from the fulcrum. The brake lever length is 1 m. The brake is to absorb a power of 15 kW at a rated speed of 720 rpm. Design the brake lever of rectangular cross section assuming the depth thrice the width and allowable stress for the lever as 80 MPa. Consider the lever as a cantilever neglecting the effect of all other forces and maximum bending moment occurs at the fulcrum. Take coefficient of friction as 0.3,

6 L2 3 2

10 L3 4 2

Unit – III

7. A C50Mn1 steel shaft having yield strength 392.3 MPa and ultimate strength 706.1 MPa transmitting 15 kW at 210 rpm is supported between two bearings 750 mm apart and has two gears keyed to it. The pinion having 24 teeth and module 6 mm is located 100 mm to the left of right bearing. This gear delivers horizontally to the right when viewed from the left bearing. The gear having 80 teeth of module 6 mm is located 150 mm to the right of left bearing and receives from a pinion in a vertical direction from below. Take C_m as 1.75 and C_t as 1.25, pressure angle 18° . Assume CCW direction from left and use allowable shear stress of 88.26 MPa.
8. Design a pair of spur gears to transmit 25 kW at 900 rpm of untreated forged steel pinion 120 mm diameter to an untreated cast steel gear which is to run at 300 rpm. The teeth are 14.5° involute. Assume medium shocks with 8-10 hours/day service. Also assume carefully cut gears with face width 10 times module.

16 L3 5 2

16 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
